

AI Crypto Trading Journal

Self-hosted · Raspberry Pi 5 · Flask 3.1 · Claude Sonnet 4.6 + Haiku 4.5 · Bitget + Blofin · SQLite WAL · 14 live data sources

842	67.3%	+\$12,840	1.49	3.82
Positions	Win rate (Long)	P&L 90d USDT	Sharpe	Calmar

* Illustrative numbers only — not real results

AI Pipeline (per trade call)

Stage	Agent	Model / Source
1 Collect	agent_data_collector	14 parallel workers
2 Candles	chart_context	CCXT · Binance Futures
3 Interpret	chart_confluence / S&R	Pure Python indicators
4 Sentiment	market_context + Nansen	REST APIs
5 Review	backtest_context	SQLite trade history
6 Score	prompt_builder → Claude	Sonnet 4.6
7 Consensus	agent_orchestrator	Gemini Flash
8 Risk	agent_risk_mgmt	Kelly criterion + ATR
9 Chart	agent_chart_draw	mplfinance PNG
10 Alert	telegram_notify	Telegram Bot API

Model Routing

Task	Model	Reason
call_analyzer	Sonnet	15-field JSON + chain-of-thought
scanner_batch	Sonnet	12 setups: entry/SL/TP reasoning
advisor	Sonnet	Full portfolio coaching
rulebook	Sonnet	Synthesises 800+ trade history
pattern_detect	Sonnet	Cross-pattern compounding
scanner_quick	Haiku	Score 0-10, runs 100× per scan
hindsight	Haiku	Binary ENTER/SKIP classification
live_trade	Haiku	Hold/Adjust/Close — latency key
trade_grader	Haiku	A–D grade, simple rubric

Claude + Gemini Consensus Scoring

Δ Claude – Gemini	Confidence	Score	Flag	Action
0 – 1 pt	High	Average	✓ Confirmed	Trade normally
2 pt	Medium	Average	~ Aligned	Trade, watch
3 pt	Low	Claude 60% + G 40%	■ Divergent	Reduce size
> 3 pt	Very Low	Claude kept	■ REVIEW	Do not trade — investigate

Data Sources — Macro → Micro

Layer	Source	Provider	Auth	Used in Pipeline
L1 Global Macro	VIX Volatility Index	CBOE · yfinance	Free	Scanner cap (>35→6.0, >25→7.5) · Confluence ×0.80 when >30
L1 Global Macro	DXY US Dollar Index	ICE · yfinance	Free	Macro regime label · Stage 3 scanner header
L1 Global Macro	Fear & Greed Index	alternative.me	Free	Scanner macro cap · Sentiment prompt · Dashboard strip
L1 Global Macro	Economic Calendar	Finnhub API	Key	FOMC/CPI/NFP → cap 7.0 when high-impact event in 24h
L1 Global Macro	BTC Dominance + Mkt Cap	CoinGecko (free)	Free	Scanner macro header · Call Analyzer market context
L2 Market Struct.	Options Skew (PCR / IV)	Deribit (free)	Free	BTC/ETH only — put/call bias, near-term IV, sentiment label
L2 Market Struct.	BTC Mempool	blockchain.com	Free	On-chain congestion context injected into prompt
L2 Market Struct.	Trending Coins (top 10)	CoinGecko (free)	Free	Is the analyzed coin trending? Prompt context block
L3 Symbol-Level	Multi-Exchange L/S Ratio	Binance+Bybit+OKX	Free	Crowd positioning · contra-signal when >65% vs trade dir.
L3 Symbol-Level	OI · Funding · Liquidations	Coinalyze API	Key	Derivatives block · funding bias · per-exchange spread
L3 Symbol-Level	Cap rank · tier · volume	CoinGecko (free)	Free	Coin context · scales Grok weight (mega 30% → micro 80%)
L3 Symbol-Level	DeFi TVL + 7d change	DefiLlama (free)	Free	DeFi tokens only — protocol health context
L3 Symbol-Level	OHLCV Candles (4H + 1D)	Binance · CCXT	Free	All indicators · S/R detection · SMT · Backtester · Charts
L4 Trade Intel.	Smart Money Wallet Flows	Nansen API	Paid	Accumulating/distributing █/█ in sentiment agent prompt
L4 Trade Intel.	Social & News Context	xAI Grok API	Key	Direction-weighted; last prompt block · lowest priority

What's New — v1.5.0

Walk-forward backtester — 70/30 temporal split on real positions; end_offset_days prevents train/test leakage

Optimizer history — Last 5 Bayesian runs stored with Sharpe + best params in Analysis tab

Scanner macro layer — VIX / F&G; / FOMC fetched once per scan; score caps applied before Stage 3 Sonnet

SMT Direction weight — +0.15 when symbol moves opposite its correlated pair (BTC↔ETH, SOL→ETH, BNB→BTC)

Calmar & Sharpe fixed — Max DD vs running peak (not ATH); sample variance N-1; wallet filter >\$1 USDT

14 data sources — CoinGecko, Deribit, Finnhub, Coinalyze, DefiLlama, Blockchain.com, Nansen, Grok all wired

Data Sources page — Tools → ███ shows all 14 sources grouped macro→micro with auth, inputs, pipeline usage

Scanner macro UI — Live VIX / F&G; / cap + macro warnings visible per scan; Gemini consensus on top-5

Technical Stack

Layer	Technology
Runtime	Python 3.13 · Flask 3.1
Database	SQLite WAL · 7-day auto backup
AI Primary	Claude Sonnet 4.6 · Haiku 4.5
AI Consensus	Google Gemini 2.0 Flash
AI Social	xAI Grok 3 Fast
Exchange	CCXT · Binance · Bybit · OKX
Charts	mplfinance · lightweight-charts
Hardware	Raspberry Pi 5 · port 8082
Tests	pytest · 351 passing

Prompt Budget Order

Priority Block		Protected until
1st	Backtest context	Always — most trade-relevant
2nd	Market data blocks	Always (Coinalyze, F&G, L/S, options...)
3rd	Rulebook	Remaining < 100 chars
4th	Chart context	Remaining < 100 chars
5th	Calibration	Remaining < 100 chars
6th	Grok social	Remaining < 150 chars · weight 0–80%